MAS · Rates of Change and Differential Equations — Patterns and Solution Approaches

Based on: marking keys + exam reports for 41 past questions from 2016–2025. Appears almost exclusively in Calculator-assumed papers. Original questions are in `topics/06_Rates_DiffEq.pdf`. Numerical examples have been verified with Python. Review status: has undergone one round of independent review and was revised according to feedback (see `Review_log.md`).

Marking-key reminders

- Motion when v is a function of x (flagged three times in the 2018, 2020 and 2022 reports): $a = dv/dt = (dv/dx)(dx/dt) = v \cdot dv/dx = d(\frac{1}{2}v^2)/dx$. dv/dx itself is not acceleration; after differentiating with respect to x, multiply by v.
- Implicit differentiation: many students are reluctant to use it (2017 report), or cannot do the second derivative (2025 report).
- Not all growth is logistic (2025 report): for equations such as $dP/dt = 0.001(100 - P)^2$, many students draw the S-shape of $rP(k - P)$ and get it wrong. Analyse the slope from the given equation itself.
- Where the slope is undefined in a slope field (2019 report): when the numerator is not 0, the solution curve has a vertical tangent there; the curve turns back at this point and is symmetric about that line.
- Rates of change must have units written (2020, 2021 reports).
- Use of the logistic equation was a difficulty flagged in the 2024 report (Q13(c), finding r and k from the explicit solution).
- Recognising standard differential equations: SHM and exponential decay (2017 report).
- Antiderivatives involving ln must include absolute values (2022 report).

Pattern 1: Implicit Differentiation (Tangents, Horizontal Points, Second Derivative)

Standard procedure

1. Differentiate both sides with respect to x: terms containing y must be multiplied by dy/dx; use the product rule for products and the chain rule for composites, e.g. $3(x^2 + y^2 - 1)^2(2x + 2y\,y')$.
2. Solve for dy/dx when the question requires it; you can also substitute the point directly to evaluate (2018A-Q20 explicitly requires you not to solve for dy/dx: "Do not attempt to obtain dy/dx as the subject").
3. Horizontal tangents: $dy/dx = 0 \Rightarrow$ numerator = 0, then solve simultaneously with the original equation to find points (2018A-Q20(c), 2021A-Q18(b)).
4. Second derivative (2025F-Q5): differentiate the once-differentiated expression again, and substitute the values of x, y, y′.
5. Derivatives of inverse trigonometric functions (2020F-Q6(b), 2024F-Q8(d)): $x = 2\tan y \Rightarrow 1 = 2\sec^2 y \cdot y' \Rightarrow$ use $1 + \tan^2 y$ to change the result into an expression in x.

Past questions: 2016A-Q13(a), 2017A-Q12(a), 2018A-Q15(a), 2018A-Q20, 2021A-Q18, 2022F-Q7(a), 2025F-Q5.

Pattern 2: Related Rates ★

Standard procedure (4–5 marks)

1. Write the relationship between the quantities (geometric or physical), establish a relationship between the variables, and eliminate unnecessary variables if needed (you can also keep two variables and differentiate implicitly with respect to t): $V = 4h^2$ (triangular prism water tank), regular hexagon area $A = (3\sqrt{3}/2)s^2$, $x = 50\tan\theta$ (lighthouse), $s(10 - h) = 30$ (shadow).
2. Differentiate with respect to t (chain rule or implicit differentiation): $dA/dt = 3\sqrt{3}\,s \cdot ds/dt$.

3. Substitute the known rate of change and the value at that time (sometimes you must find θ or h first).
4. Include units. Example: $s = 4$, $ds/dt = 0.5 \Rightarrow dA/dt = 6\sqrt{3} \approx 10.39$ cm²/s (2022A-Q8).
5. Lighthouse type (2021A-Q9): 3 revolutions per minute ⇒ $d\theta/dt = 6\pi$ rad/min $= \pi/10$ rad/s; when $x = 100$, $\sec^2\theta = 1 + (100/50)^2 = 5$.

Past questions: 2016S-A11(a), 2016A-Q15, 2017A-Q18(c), 2019A-Q18(c), 2020A-Q20(b), 2021A-Q9, 2022A-Q8, 2023A-Q19, 2025A-Q14(b). (2023A-Q17(c) is an increments formula question; see Pattern 8.)

Related rates in circular motion (Ferris wheel, carousel)

Approach

1. Angular speed: one revolution is 2π radians and takes T seconds ⇒ $d\theta/dt = 2\pi/T$ (1 mark for "show that").
2. Model using geometry: cosine rule $s^2 = 89 - 80\cos\theta$ (2017A-Q18(b)), or height relative to the centre $y = 80\sin(\theta - \pi/2)$ (2019A-Q18(b); explain the value of the phase α; height above the ground = y + 80, so 100 m above the ground corresponds to $y = 20$).
3. Differentiate implicitly with respect to t: $2s \cdot ds/dt = 80\sin\theta \cdot d\theta/dt$.
4. Find the maximum rate of change: differentiate again and set it to 0; you can also interpret it geometrically: ds/dt is greatest when PC is tangent to the circle.

Past questions: 2017A-Q18, 2019A-Q18.

Pattern 3: Slope Fields

Common question types and approaches

- Find the slope at a point: substitute directly (1 mark).
- Infer the equation from a slope field: observe whether the slope changes only with x (segments on the same vertical line are parallel) or only with y; where the slope is 0; where the slope is undefined. For example, 2016A-Q18: $dy/dx = a - bx$ ($a, b > 0$); one condition is read from the diagram (slope is 0 at $x = 2$) and one is given in the question (slope is 0.5 at A(1, 2)) ⇒ $dy/dx = 1 - 0.5x$. The mark point in (a) requires stating that the intercept is positive and the slope is negative.
- Draw solution curves: pass through the specified point, follow the direction of the short line segments, and indicate symmetry; where the slope is undefined the tangent is vertical (2019A-Q11(b), 2021A-Q10(c)); where the slope is 0 there is a horizontal tangent; on the line where the slope is greatest, the curve has a point of inflection. 2025A-Q16: $dy/dx = (x - 1)/((y + 1)^2 + 1)$, $x = 1$ is the minimum point, and the point of inflection is at (3, −1) (the slope is greatest on $y = -1$); the key awards 1 mark.
- Analytical solution: solve by separating the variables (see Pattern 4).

Past questions: 2016A-Q18, 2016S-A21, 2017A-Q11, 2019A-Q11, 2021A-Q10, 2023A-Q11, 2025A-Q16.

Pattern 4: Separation of Variables

Standard procedure (3–4 marks)

1. Write as $g(y)\,dy = f(x)\,dx$, put integral signs on both sides (the exact wording in the key for 2021A-Q10(b) and 2022A-Q17(d) is "separates the variables as an integration statement"; in other years it says "separates the variables correctly").
2. Integrate both sides, adding only one constant; ln must include absolute values.
3. Substitute the initial condition to find the constant.
4. Write in explicit form y = … as required.

Common models

- $dP/dt = kP \Rightarrow P = P_0e^{kt}$ (2019A-Q17(a): the instantaneous growth rate is 10% of the population, i.e. $dP/dt = 0.1P$; from $P(45) = 30000$, work back to 1963: $P_0 = 30000e^{-4.5} \approx 333$. Do not treat it as "10%

growth per year" and calculate using 1.1^t; that gives about 412).

- $dA/dt = -0.4A \Rightarrow A = 8e^{-0.4t}$ (2017A-Q17(c)).
- $dh/dt = -1/(100h) \Rightarrow h^2 = h_0^2 - t/50$ (2016A-Q15(d)).
- $dv/dt = -9.8 - 2v \Rightarrow v = 4.9(e^{-2t} - 1)$; depth must be found by integrating v (2024A-Q17).
- $dV/dt = 300e^{-V/12000} \Rightarrow$ separation gives $e^{V/12000}dV = 300\,dt$ (2022A-Q17(d)).

Past questions: 2016A-Q15, 2016S-F3, 2017A-Q11(c), 2019A-Q11(c), 2021A-Q10(b), 2022A-Q17(d), 2023A-Q11(b), 2024A-Q17(c), 2025A-Q17(c).

Pattern 5: Logistic Growth ★

Formulas: $dP/dt = rP(k - P) \Leftrightarrow P(t) = k / (1 + Ae^{-rkt})$.

- Limiting value (carrying capacity) = k ($t \to \infty$);
- Growth is fastest when $P = k/2$ (point of inflection), maximum growth rate = $r(k/2)^2$;
- Find r and k from the explicit solution: for example, $P = 18000/(10.25e^{-0.15t} + 1) \Rightarrow k = 18000$, $rk = 0.15$. If the question writes $dP/dt = (1/r)P(k - P)$, then $1/r = 0.15/18000 \Rightarrow r = 120000$; maximum growth rate = $0.15 \times 9000 \times ½ =$ 675 horses/year (2024A-Q13, consistent with the key).

Common question types

- Find the proportionality constant from the initial rate of change (in 2018A-Q18(a) the question calls it k: $60 = k \cdot 100 \cdot 1500 \Rightarrow k = 0.0004$; the carrying capacity is 1600). Use the letters given in the question: carrying capacity = the non-zero P that makes $dP/dt = 0$.
- Increment estimate: $\delta P \approx (dP/dt) \cdot \delta t$; note the time units (1 month = 1/12 year, 15 minutes = 0.25 hours).
- Use partial fractions to separate the variables and derive the explicit solution (2016S-A20, 2018A-Q18(c)).
- Explain the growth: "When $P < k/2$, the growth rate increases as P increases; when $P > k/2$, the growth rate decreases and P approaches k" (2022A-Q16(a)).
- Sketch after the carrying capacity changes: new point of inflection, new horizontal asymptote (2022A-Q16(b)); when $k/2 < P_0 < k$, the curve is concave down throughout, increases towards k, and has no point of inflection (2020A-Q19(e): $P_0 = 1.3$, $k = 2.4$); when $P_0 > k$, the curve decreases and approaches k.
- Non-logistic (2025A-Q17): $dP/dt = 0.001(100 - P)^2 \Rightarrow$ the slope decreases monotonically as P increases, there is no point of inflection, and the asymptote is $P = 100$.

Past questions: 2016S-A20, 2018A-Q18, 2019A-Q17(b)–(d), 2020A-Q19, 2022A-Q16, 2024A-Q13, 2025A-Q17.

Pattern 6: Velocity as a function of displacement (v = f(x)) ★★ Report focus

Standard procedure

1. Acceleration: $a = v \cdot dv/dx$ (or $d(½v^2)/dx$).
   - $v = -0.2x \Rightarrow a = (-0.2x)(-0.2) = 0.04x$;
   - $v = x/8 \Rightarrow a = (x/8)(1/8) = x/64$.
2. Find x(t): $dx/dt = v(x) \Rightarrow$ separate variables $\Rightarrow$ substitute the initial position.
   - $v = x/8$, $x(0) = 192 \Rightarrow x = 192e^{t/8}$, $v(0) = 24$, consistent with the given condition $\Rightarrow$ at 11 seconds, velocity is $24e^{11/8} \approx 94.92$ m/s, distance travelled is $192(e^{11/8} - 1) \approx 567.37$ m (2022A-Q11).
   - $v = -0.2x$, $x(0) = 4 \Rightarrow x = 4e^{-0.2t} \Rightarrow$ when $x = 2$, $t = 5 \ln 2 \approx 3.47$ s (2020A-Q14).
3. Determine whether it is SHM: It is SHM only if a is proportional to the displacement (relative to the

equilibrium point) and in the opposite direction, i.e. $a = -n^2(x - c)$. $a = 0.04x$ has a positive coefficient, so it is not (2020A-Q14(b)).

Past questions: 2018A-Q13, 2020A-Q14, 2022A-Q11.

Pattern 7: Simple harmonic motion (SHM)

Formulas: $\ddot{x} = -n^2x \Rightarrow x = A\cos(nt + \varphi)$ or $A\sin(nt + \varphi)$; period $T = 2\pi/n$; maximum speed $= An$; $v^2 = n^2(A^2 - x^2)$.

Common question types

- Determine A and phase from the initial conditions: $x(0) = 0$ and $v(0) > 0 \Rightarrow x = A\sin(nt)$ (2017A-Q17). 2024A-Q12 only says it starts from the equilibrium position and does not give a direction; $\pm 25\sin(\pi t/2)$ is accepted either way (total vertical range 50 cm $\Rightarrow A = 25$).
- Find the amplitude from $v^2 = n^2(A^2 - x^2)$: $n = 2$, $x(0) = -1$, $v(0)^2 = 32 \Rightarrow 32 = 4(A^2 - 1) \Rightarrow A = 3$, $T = \pi$ (2025A-Q15).
- Distance travelled: each period covers 4A, each half period covers 2A. 2025A-Q15(b): 10π seconds = 10 periods $\Rightarrow 4 \times 3 \times 10 = 120$ cm. 2017A-Q17(b): $T = 2$, 5 seconds = 2.5 periods $\Rightarrow 10A = 80$ cm. If less than half a period and the starting point is not at the equilibrium position or an endpoint, use $\int |v|\, dt$.
- Amplitude decay (damping): $dA/dt = -kA \Rightarrow A = A_0e^{-kt}$; the condition for appearing to stop is $A < 0.01$ (2017A-Q17(d)).

Past questions: 2016S-A15, 2017A-Q17, 2019A-Q18(d) (prove that the Ferris wheel's $y(t)$ satisfies $d^2y/dt^2 = -(\pi/36)^2y$), 2021A-Q12, 2024A-Q12, 2025A-Q15.

Pattern 8: Incremental formula (MAS version)

$\delta y \approx (dy/dx)\delta x$, also applies to differential equations: $\delta N \approx (dN/dt)\cdot\delta t$, where dN/dt is given directly by the equation (2018A-Q18(b), 2020A-Q19(c), 2025A-Q17(b)); $\delta h \approx \delta V / (dV/dh)$ (2023A-Q17(c)). Past questions: 2016A-Q11(b), 2018A-Q18(b), 2020A-Q19(c), 2022A-Q17(b), 2023A-Q17(c), 2025A-Q17(b).